

MacSync-WGSX2 Wi-Fi H₂S & NH₃ Gas Sensor Datasheet

Product Overview

MacSync-WGSX2 is a high-precision **Wi-Fi-based quantitative H₂S & NH₃ gas monitoring sensor** designed for real-time measurement of **hydrogen sulfide (H₂S)** and **ammonia (NH₃)** concentrations in ppm for industrial safety, environmental compliance, and occupational health applications within Wi-Fi-enabled environments.

Built using **industrial-grade electrochemical gas sensors**, the device delivers accurate, stable, and repeatable ppm-level measurements with fast response times, enabling early detection of hazardous gas buildup and proactive safety intervention. Unlike qualitative odor sensors, MacSync-WGSX2 provides **true quantitative gas concentration data**, making it suitable for regulatory monitoring, safety audits, and process control.

MacSync-WGSX2 communicates directly over **standard Wi-Fi networks (2.4 GHz / 5 GHz)** to **on-premise servers, SCADA systems, safety dashboards, or cloud platforms** using industry-standard protocols such as **MQTT, HTTP, and HTTPS**, without the need for gateways or long-range wireless infrastructure.

Housed in a **rugged IP-rated enclosure**, the sensor is suitable for deployment in **sewage treatment plants (STP), wastewater facilities (WWTP), chemical plants, factories, confined indoor spaces, utility rooms, and industrial work zones** where reliable Wi-Fi connectivity is available. **BLE-based configuration via the Macnman Maya App** enables quick provisioning, diagnostics, calibration checks, and maintenance.

Whether deployed for **worker safety, toxic gas exposure monitoring, ventilation control, or compliance reporting**, **MacSync-WGSX2 delivers reliable, real-time Wi-Fi-based toxic gas monitoring**, transforming hazardous gas risks into actionable data.



Key Features

Dual-Gas Quantitative Measurement (H_2S + NH_3)

Real-time ppm-level measurement of hydrogen sulfide (H_2S) and ammonia (NH_3) for safety-critical and compliance applications.

Wi-Fi Based Connectivity

Direct IP communication over standard Wi-Fi networks (2.4 GHz / 5 GHz) without gateways or long-range wireless infrastructure.

Industrial & Corrosion-Resistant Design

Rugged IP-rated enclosure engineered for gas-rich, corrosive environments such as STP, WWTP, factories, and utility areas.

Industrial-Grade Electrochemical Sensors

High sensitivity, fast response, and stable electrochemical sensing elements ensure accurate and repeatable gas concentration readings.

Real-Time Gas Alerts & Threshold Events

Configurable alarms for toxic gas exposure, ventilation control, and worker safety interventions.

Battery-Operated & Externally Powered Variants

Available in battery-powered and 9–36 V DC powered versions to suit both fixed and flexible indoor deployments.

Standard IoT Protocol Support

Supports MQTT, HTTP, and HTTPS for seamless integration with SCADA systems, safety dashboards, and cloud platforms.

BLE Configuration via Maya App

Local provisioning, diagnostics, calibration checks, and configuration using the Macnman Maya mobile application.

Suited Applications

Sewage & Wastewater Treatment Plants (STP / WWTP)

Continuous ppm-level monitoring of H₂S and NH₃ near inlet chambers, aeration tanks, sludge handling areas, and pumping stations within Wi-Fi-enabled facilities.

Chemical & Process Industries

Quantitative gas monitoring around reactors, storage areas, and handling zones for early leak detection and process safety.

Livestock & Poultry Facilities

Ammonia (NH₃) concentration monitoring to protect animal health, improve ventilation efficiency, and maintain welfare standards.

Environmental Compliance & Safety Audits

Supports regulatory monitoring programs, safety audits, and data-driven reporting using accurate ppm-level gas measurements.

Secure Wi-Fi Communication

WPA2 / WPA3 authentication with TLS-encrypted data transmission for secure IP-based communication.

Designed for Safety & Compliance

Suitable for occupational safety monitoring, toxic gas exposure control, and regulatory reporting requirements.

Industrial Safety & Worker Protection

Real-time toxic gas detection in factories, process plants, and production floors to prevent hazardous exposure and ensure occupational safety compliance.

Confined Indoor Spaces

Monitoring H₂S and NH₃ buildup in utility rooms, basements, pump rooms, tunnels, and enclosed work zones with reliable Wi-Fi coverage.

Industrial Warehouses & Storage Areas

Detection of toxic and corrosive gas accumulation in chemical storage rooms, fertilizer warehouses, and logistics facilities.

Ventilation Control & Automation

Enables automated ventilation triggers and alarms based on real-time H₂S and NH₃ concentration thresholds.

Technical Specifications

General Specifications

Part Order Number	MacSync-WGSX2
Product Brand	Macnman Technologies Pvt Ltd

NH₃ Measurement

Measurement Range	0 – 100 ppm
Max Detecting Concentration	200 ppm
Sensitivity	(50–100) nA/ppm
Resolution	0.5 ppm
Repeatability	< 10% of output value
Stability (per month)	< 10%
Zero Drift	-3 to +10 ppm (at -20°C to +40°C)
Expected Lifetime	2 years @ 2 ppm NH ₃
Sensor Type	Electrochemical NH ₃ gas sensor
Calibration	Factory calibrated

H₂S Measurement

Measurement Range	0 – 100 ppm
Max Detecting Concentration	500 ppm
Sensitivity	(0.8 ± 0.15) µA/ppm
Resolution	0.1 ppm
Repeatability	< 2% of output value
Stability (per month)	< 2%
Zero Drift	≤ 0.2 ppm
Expected Lifetime	2 years (in clean air)
Sensor Type	Electrochemical H ₂ S gas sensor
Calibration	Factory calibrated

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client) Access Point Mode (Configuration)
Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3

	TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSync is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers Private Cloud Deployments Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud (via MQTT / HTTPS) Custom IoT Platforms
Integration Methods	MQTT Brokers REST / HTTPS APIs Webhooks
Enterprise Systems	SCADA Systems BMS / IBMS Platforms Data Lakes & Analytics Engines

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass field Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode ~150 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacSync-WGSX2 Variants

VARIANT NAME	GAS TYPE	POWER SOURCE	BATTERY CAPACITY	IDEAL DEPLOYMENT
MacSync-WGSX2-PO	H ₂ S + NH ₃	External Power (9–36 V DC)		Industrial sites with stable power — refineries, chemical plants & smart-city nodes
MacSync-WGSX2-BO	H ₂ S + NH ₃	Battery Powered	19 Ah	Remote sewer lines, pumping stations, livestock farms & outdoor odor monitoring

What's in the Box?

Make sure box contains following items before installations



MacSync



Mounting
Clip



User manual &
Warranty Card



Junction
Box*



Mounting
Screws

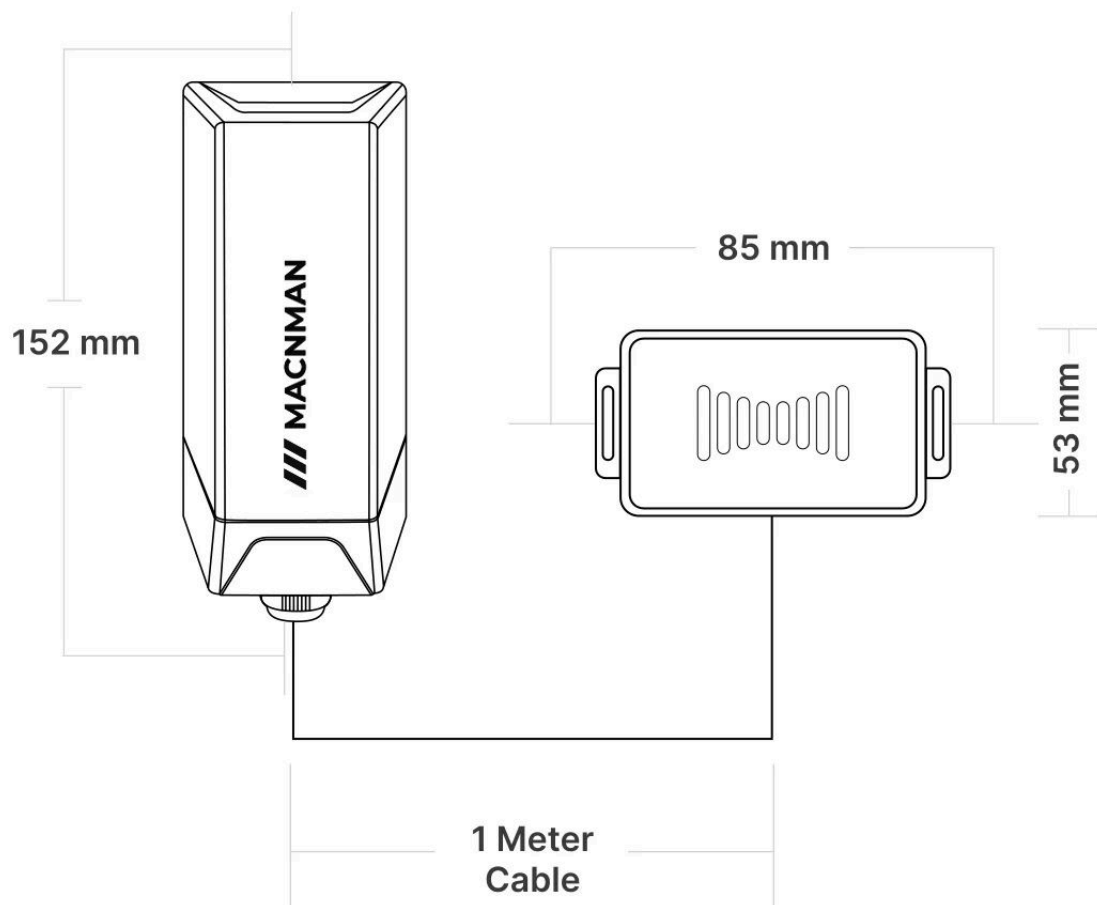


Power
Adapters*

i Note

Accessories must be purchased separately & may not be included in the box.

MacSync-WGSX2 Dimensions



Frequently Asked Questions

What is the MacSync-WGSX2 used for?

MacSync-WGSX2 is a high-precision Wi-Fi quantitative gas sensor for real-time ppm measurement of hydrogen sulfide (H₂S) and ammonia (NH₃). Typical applications include Sewage & Wastewater Treatment Plants (STP / WWTP), Industrial Safety & Worker Protection, Chemical & Process Industries, Confined Indoor Spaces, Livestock & Poultry Facilities and Industrial Warehouses & Storage Areas.

What does the MacSync-WGSX2 measure, and how accurately?

Measurement specifications of the MacSync-WGSX2:

- NH₃: range 0 – 100 ppm, resolution 0.5 ppm.
- H₂S: range 0 – 100 ppm, resolution 0.1 ppm.

How does the MacSync-WGSX2 connect and send data over Wi-Fi?

The MacSync-WGSX2 connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band).

Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

How is the MacSync-WGSX2 powered?

The MacSync-WGSX2 is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacSync-WGSX2?

The MacSync-WGSX2 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacSync-WGSX2 configured?

The MacSync-WGSX2 is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MacSync-WGSX2 work with?

The MacSync-WGSX2 can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- Related datasheets: [MacSync L Odor X2 LoRaWAN NH3 and H2S odor sensor](#), [MacSync-W-Odor-X1 Wi-Fi qualitative odor sensor](#), [MacSync-WCO2X Wi-Fi CO2 indoor air quality sensor](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)